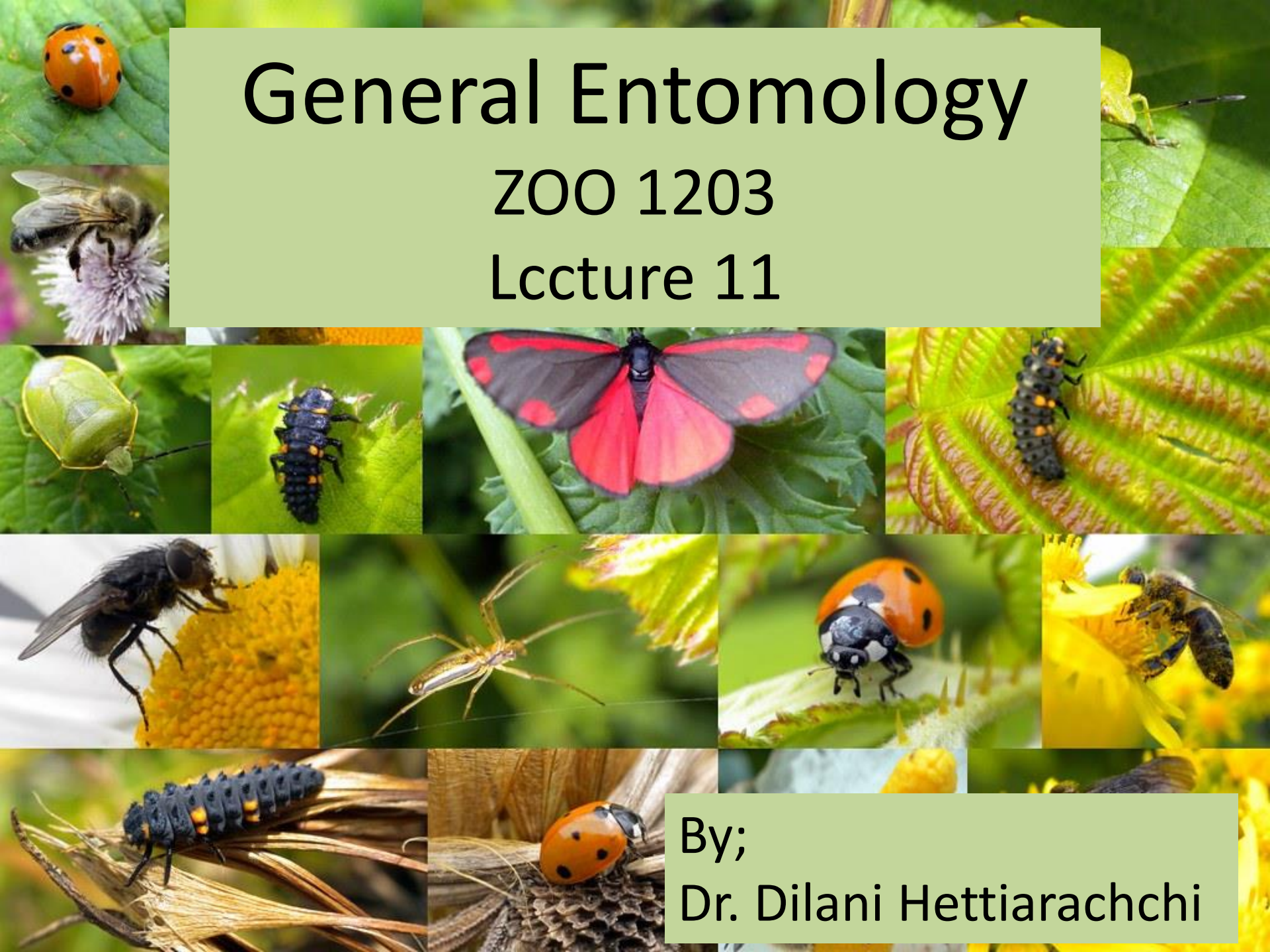


A collage of various insects is arranged around a central text box. The insects include a ladybug on a green leaf (top left), a bee on a purple flower (middle left), a green shield bug on a leaf (bottom left), a black and yellow caterpillar on a green leaf (middle left), a red and black moth on a green leaf (center), a black and yellow caterpillar on a green leaf (middle right), a fly on a yellow flower (bottom left), a long-legged insect on a yellow flower (bottom center), a ladybug on a green leaf (bottom right), and a bee on a yellow flower (bottom right). The central text box is light green and contains the title 'General Entomology', the course code 'ZOO 1203', and the lecture number 'Lccture 11'. The bottom right corner of the collage features a light green box with the text 'By; Dr. Dilani Hettiarachchi'.A collage of various insects is arranged around a central text box. The insects include a ladybug on a green leaf (top left), a bee on a purple flower (middle left), a green shield bug on a leaf (bottom left), a black and yellow caterpillar on a green leaf (middle left), a red and black moth on a green leaf (center), a black and yellow caterpillar on a green leaf (middle right), a fly on a yellow flower (bottom left), a long-legged insect on a yellow flower (bottom center), a ladybug on a green leaf (bottom right), and a bee on a yellow flower (bottom right). The central text box is light green and contains the title 'General Entomology', the course code 'ZOO 1203', and the lecture number 'Lccture 11'. The bottom right corner of the collage features a light green box with the text 'By; Dr. Dilani Hettiarachchi'.

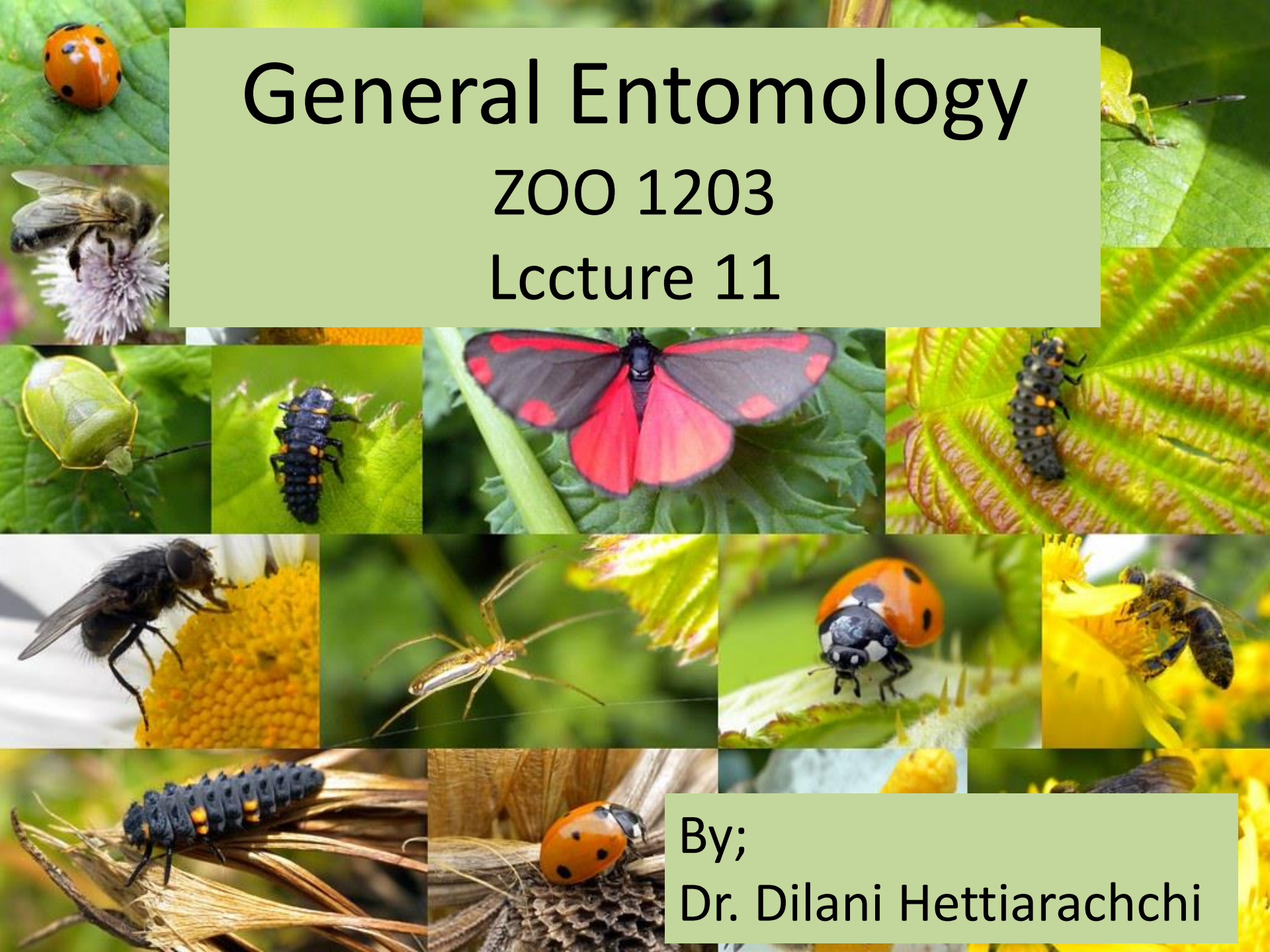
General Entomology

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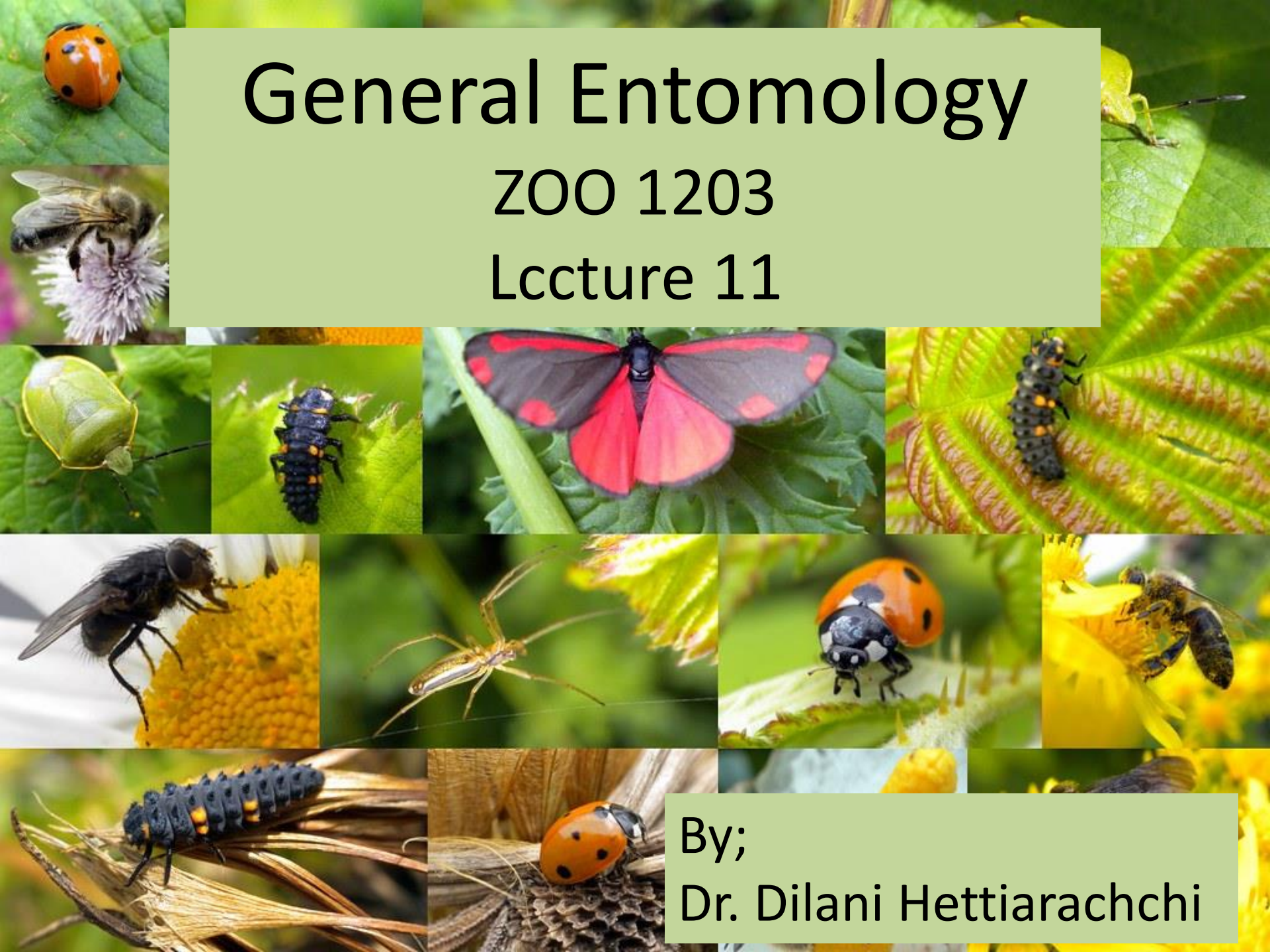
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General Entomology

ZOO 1203

Lccture 11

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Lesson Objectives

After completion you should be able to;

- Explain the evolutionary involvement of insects

Evolution of insects

Phylogenetic

- Various groups (taxa), especially the orders, are fairly well defined
- But the phylogenetic relationship among taxa are in conjecture even at the level of order
- Ex: the beetles (Coleoptera) form a discrete group (having fore wings modified as elytra) and they have no close relatives.
- Stonefly (Plecoptera) and mayfly (Ephemeroptera) resemble each other closely but it is not the case phylogenetically

- Stoneflies → more closely related to the orthopteroids (cockroaches, mantids, termites, earwigs, grasshoppers and crickets)

Stoneflies



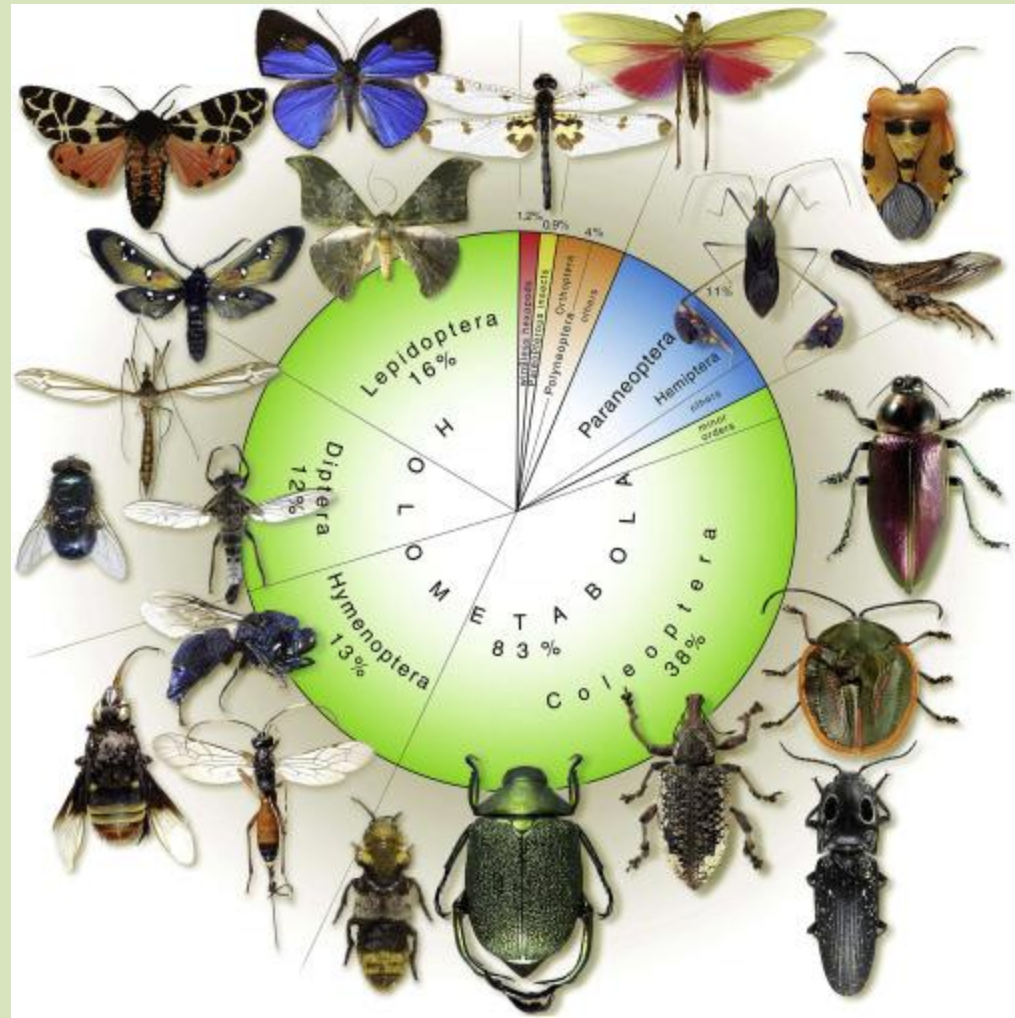
Mayflies



- Resemblance need not indicate evolutionary relationship
- Only similarity as a result of common ancestry (homology) can provide information regarding phylogeny
- Current taxonomy of insects are mostly based on groups (taxa) with distinct morphology
- These include monophyletic as well as paraphyletic
- But unlikely to have any higher level polyphyletic

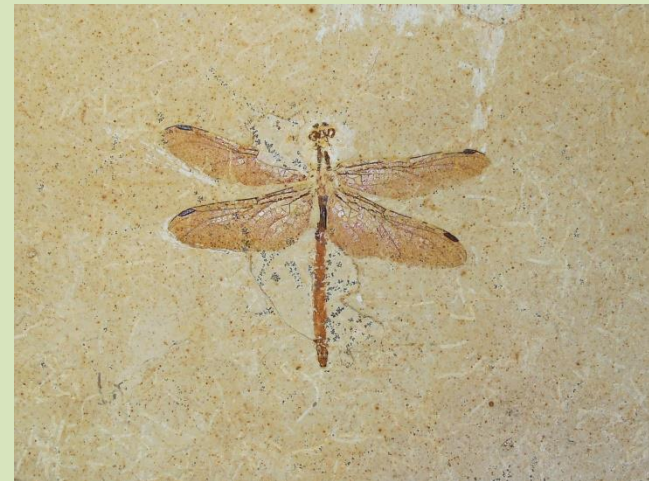
(refer evolution notes on paraphyletics, monophyletics and polyphyletics)

- Insects were among the first animals to colonize and exploit terrestrial and freshwater ecosystems.
- Insects were the first to master flights
- And also they were the first to establish social societies



The insect fossil records

- The problem with fossils is they are rare.
- When you have tiny squishy animals involved, they are even less common.
- This new research uses time estimates based on geological evidence from fossils in combination with estimated times of divergence based on molecular evidence.



- The oldest fossil hexapods known from 394 – 401 mya in the lower Devonian of Scotland

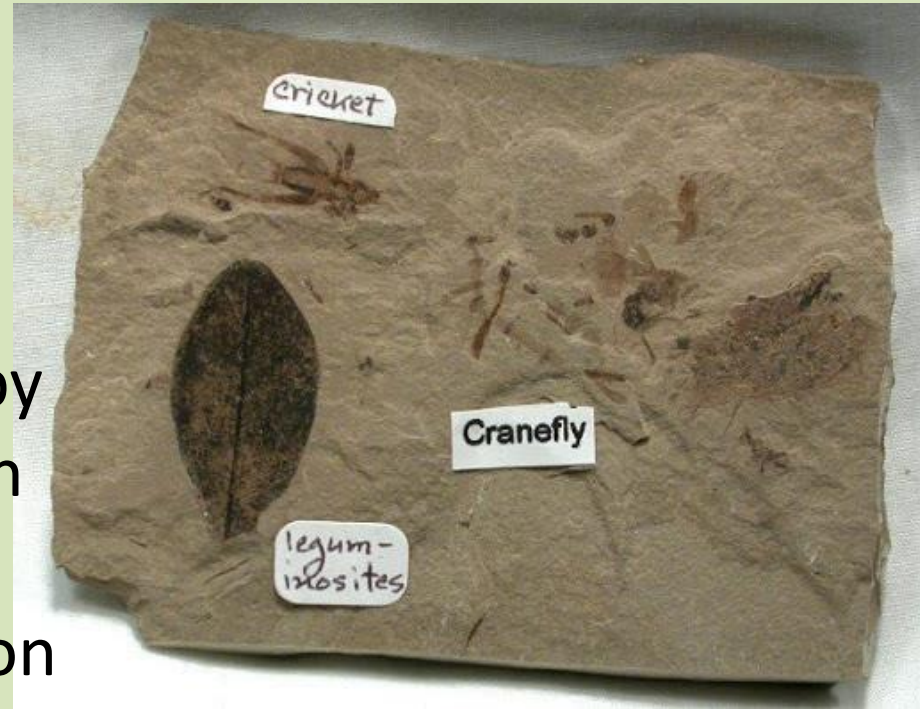


- Thus can hypothesis that insects originated in the Late Silurian with the earliest terrestrial ecosystems
- Arthropoda, whose origin is thus far unknown, probably arose in Precambrian times, perhaps as many as 1 billion years ago.
- Earliest fossil of ectognathus (class of wingless and ametabolous arthropods) are found in younger Devonian deposits of northeastern North America

- There are lower Devonian fossil plants showing damage resembling that caused by the piercing and sucking mouth parts of insects and mites
- Evidence of early Insecta or their relatives are unlikely to be found as appropriate freshwater fossils are scarce
- Ephemeroptera, Blattodea, Orthoptera and some hemipteroids fossils were found dating back to Carboniferous era
- Beak-like piercing mouth parts and expanded clypeus of some carboniferous insects indicates an early origin of feeding on plants

- Gymnosperms became more abundant during the Permian era
- Earlier it was dominated by ferns
- There was a dramatic increase in insect ordinal diversity with some 30 orders known from the Permian
- Some were extinct
- The plant sucking Hemipterans may have been associated with the newly available plants with thin cortex and sub-cortical phloem
- Other Permian insects include pollen eaters

- A notable feature of some Carboniferous and Permian insects was their Large Size
- There are fossils of Ephemeroptera and dragonfly-like Meganisoptera with wingspan up to 45 cm and 71 cm respectively
- On explanation for this gigantism is that respiration restrictions on insect size might have been alleviated by elevated atmospheric oxygen levels which would have promoted greater O₂ diffusion in the trachea



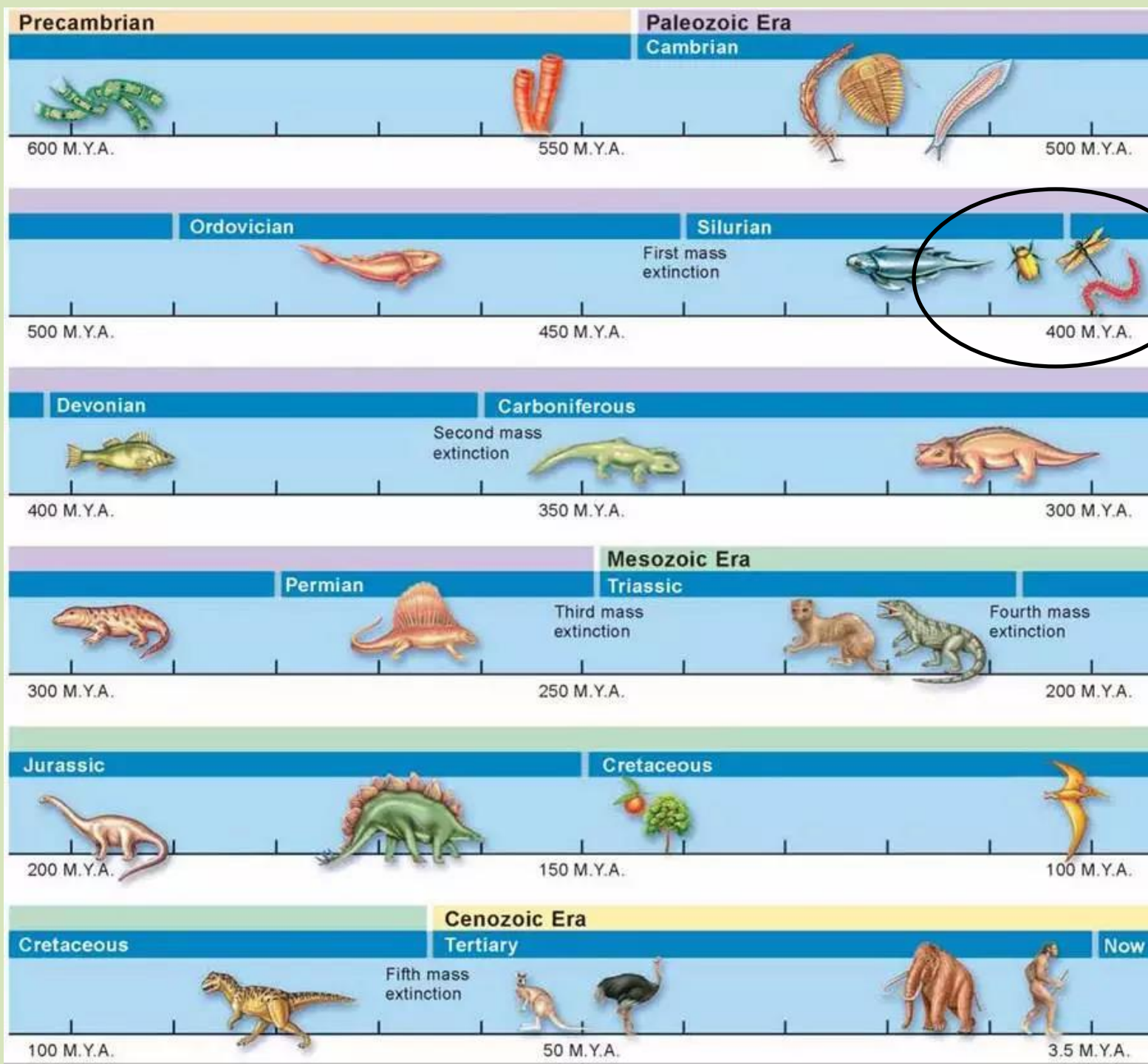
- But it is a hypothesis
- Physiological and morphological consequences of alteration of gaseous composition have been little studied in insects
- Thus except, hymenoptera and Lepidoptera, major orders of modern insects appear to have been well established by the beginning of the Triassic (about 245 Mya)

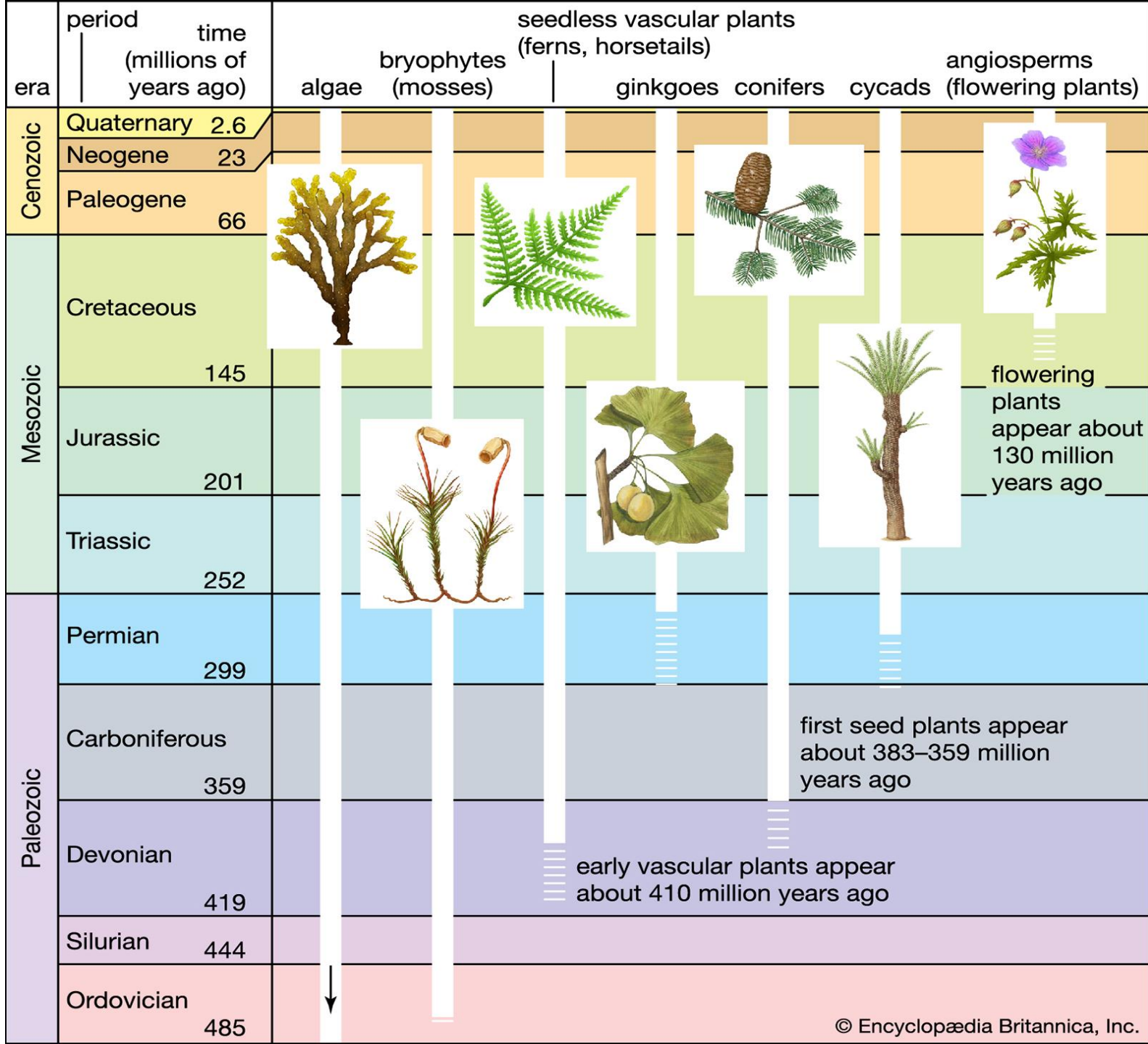


- Fossil evidence indicate that the numerical dominance of insects coincide with the angiosperm (flowering plant) diversification in the Cretaceous (145 – 65 Mya)
- But the rate of appearing of new insect families slowed during the Cretaceous
- Most of the major mouth part types seen in insects have evolved prior to the angiosperm radiation
- Perhaps it happens due to earlier plant radiations
- By the end of Cretaceous the fossil insects fauna looks fairly modern

- Tertiary period (65 – 1.6 Mya) excellent arthropod specimens have been preserved in amber
- Specimens were preserved well in amber when compared with compressed fossils
- There have been a number of attempts to extract, amplify and sequence ancient DNA from fossil insects preserved in amber (idea in the movie Jurassic Park)
- The amber resin is thought to dehydrate specimens and protect their DNA from bacterial degradation
- However authentication of the extracted DNA sequences failed



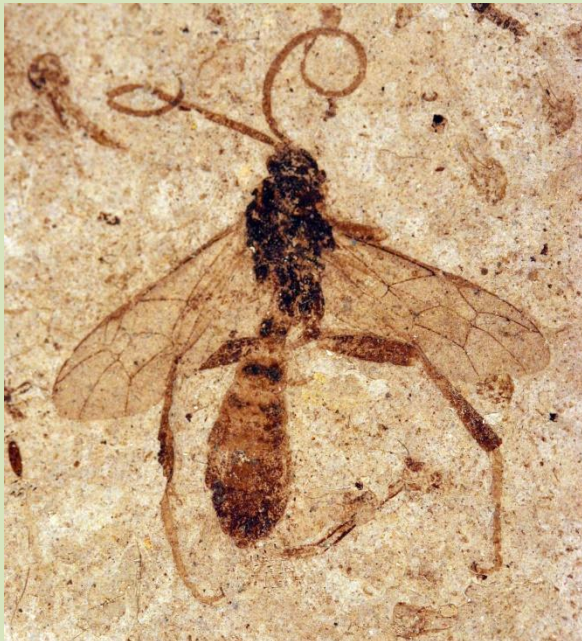
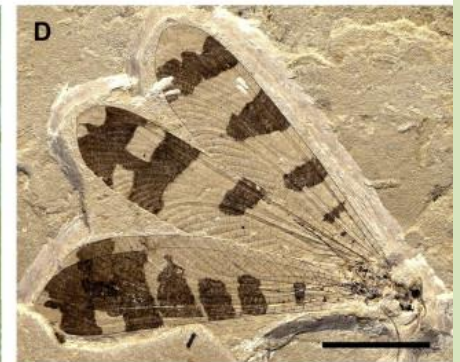
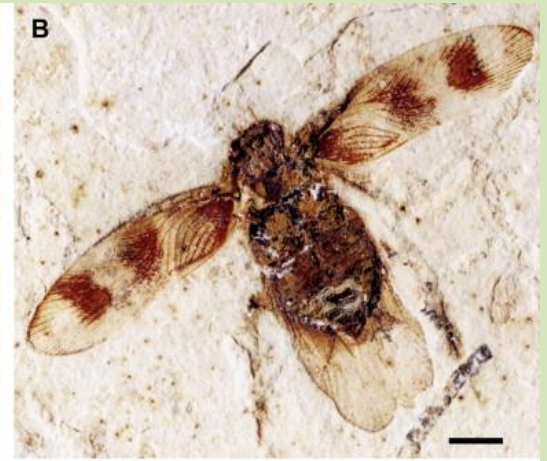
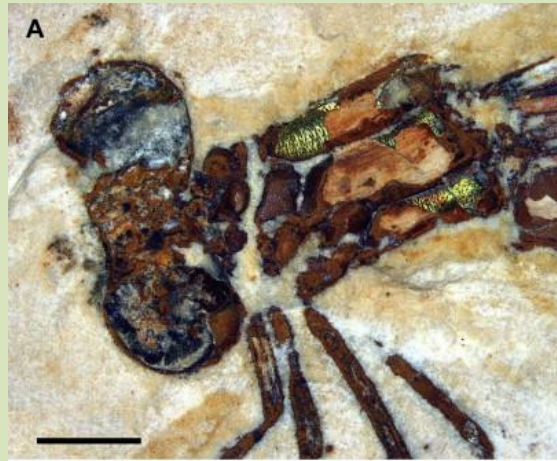




- Many species and higher taxa are much older than was previously thought
- The latest phylogenomic studies revealed that origin of insects (hexapods) dated back to Ordovician time (~479Ma)
- radiation of ectognathous insects in the Early Silurian (~441Ma)
- These estimates imply that insects colonized land at roughly the same time as plants

(Read the paper; Phylogenomics resolves the timing and pattern of insect evolution given with the PPT)

- Palaeontology has much to bring to studies on insect evolution



THANK YOU!

